

Medical Prescription Recognition for Efficient Healthcare Workflows

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Outline

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Introduction

- **Medication errors from illegible prescriptions are a critical healthcare challenge**
- WHO estimates that 1 death per million population occurs annually ($\approx 163,000$ deaths in EU) [1]
- **97.1%** of doctors in developing countries still use handwritten prescriptions [1]
- **Key challenges:**
 - Mixed content documents (printed + handwritten) [4]
 - Specialized medical terminology [7]
 - Multilingual requirements [9]
 - Time-critical processing needs [5]

Problem Statement

Handwritten medical prescriptions are difficult to digitize due to inconsistent handwriting, mixed printed and handwritten regions, and frequent image quality issues. These challenges make it hard to accurately extract medication details and can lead to serious dispensing errors, as highlighted by Bhuiyan et al. [1]. This project aims to develop a reliable end-to-end system that can segment prescription regions, recognize handwritten text, and validate extracted information to support safer and more efficient healthcare workflows.

Tabassum et al. (2022), Nature Scientific Reports [11]

- Created 'Handwritten Medical Term Corpus' with 17,431 samples
- Introduced RSS (Rotate-Shift-Stretch) data augmentation
 - Expanded dataset to 1,591,100 samples
 - Represents real-world handwriting diversity
- Bidirectional LSTM architecture
- 93.0% average accuracy (94.5% max)
- 19.6% improvement over non-augmented approaches

Literature Review Contd..

Pavithiran et al. (2024) [9]

- Hybrid CNN-RNN-LSTM architecture
- Unicode support for multilingual prescriptions
- 89% accuracy on multi-language datasets

Utkarsh Shaw et al., 2021) [10]

- Three-layer neural network using EMNIST dataset
- 94.49% training accuracy, 80.72% test accuracy
- Segmentation of handwritten strokes into individual characters

Literature Review Contd..

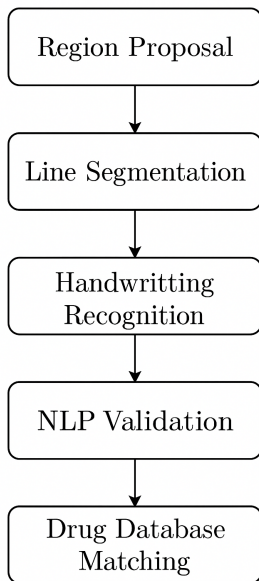
Dhar et al. (2020), Springer Multimedia Tools and Applications [2]

- Novel method for separating printed and handwritten text in prescriptions
- Uses gradient-based features and ensemble classifiers
- Addresses critical preprocessing challenge for prescription analysis
- Achieved 97.8% classification accuracy on mixed-content documents

Fajardo et al. (2019), IEEE HNICEM [3]

- Deep learning approach for cursive handwriting in medical contexts
- Custom CNN architecture optimized for doctor's handwriting patterns
- Demonstrated effectiveness on Philippine prescription dataset
- Incorporated specialized preprocessing for varying ink intensities

System Architecture



System Overview

Five-Stage Pipeline:

- 1 Preprocessing (deskew, privacy filtering)
- 2 Region Segmentation using YOLOv8-seg [6]
- 3 Line Segmentation (OpenCV-based)
- 4 Handwriting Recognition using TrOCR [8]
- 5 NLP Validation with BioBERT + Drug Lexicon Matching [7]

Data Acquisition and Preparation

Datasets:

- IAM OnDB + HMT handwriting corpora
- HuggingFace prescription datasets
- IEEE Printed/Handwritten Text Dataset
- **Custom dataset:** Indian prescriptions (pharmacy collection, annotated via CVAT)

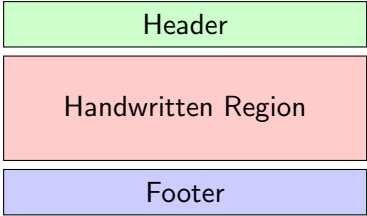
Augmentation:

- Rotation ($\pm 5-10^\circ$), translation, scaling
- Elastic transforms, brightness/contrast jitter
- Expands limited line-level dataset for OCR

Region Segmentation Module

YOLOv8 Segmentation [6]:

- Detects and segments key zones:
 - Header (Doctor/Clinic Details)
 - Handwritten Region (Drugs)
 - Footer (Signature, Notes)
- Trained on manually annotated data via CVAT
- Outputs bounding masks used to crop handwritten sections
- Faster, lightweight alternative to traditional Mask R-CNN



Header

Handwritten Region

Footer

Handwriting Recognition Pipeline

TrOCR (Transformer OCR):

- Pretrained vision encoder (ResNet/ViT) + Transformer decoder
- Fine-tuned on IAM, HMT, and custom line-level data
- Handles longer dependencies better than BiLSTM
- Improved accuracy on real-world prescription handwriting

Workflow:

- 1 Segment lines from handwritten zone
- 2 Feed line crops to TrOCR model
- 3 Output: transcribed drug names + dosage

NLP Validation Module

BioBERT for NER:

- Identifies drug names, dosage units, and frequency expressions
- Robust to noisy OCR outputs

Drug Lexicon Matching:

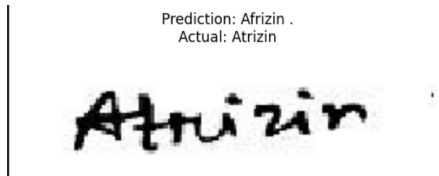
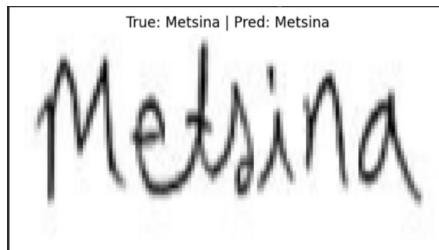
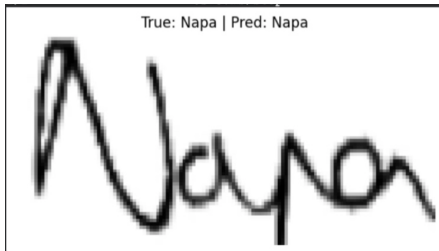
- Validates predictions using RxNorm + Indian trade names
- Fuzzy matching (Levenshtein) for OCR corrections

Dosage Pattern Extraction:

- Regex-based parsing of structured dosage (e.g., 500 mg 1-0-1)

Experimental Results

Recognition Output:



Evaluation Metrics

Phase 4 OCR Module (Medicine Name Recognition)

Model: microsoft/trocr-base-handwritten (fine-tuned)

Dataset: Doctor's Handwritten Prescription BD (Testing split, $n = 780$)

Metric	Value
Exact-Match Accuracy	96.28%
Character Error Rate (CER)	2.87%
Word Error Rate (WER)	3.92%
Total Samples	780

Work Done

- Collected prescriptions from Indian pharmacies.
- Annotated using CVAT.
- Fine-tuned TrOCR (`trocr-base-handwritten`) for handwritten medicine-name recognition

Novelty in Approach

- **Lightweight Layout Segmentation:** Propose using YOLOv8-seg for fast segmentation of prescription regions instead of heavier models like Mask R-CNN.
- **Transformer-based Handwriting Recognition:** Uses TrOCR as a modern alternative to RNNs for recognizing diverse handwritten medical text.
- **Targeted RSS Augmentation:** Plan to apply Rotation, Shifting, and Stretching on line-level handwritten data to improve generalization.
- **Domain-Specific NER and Validation:** Integrate medical NER (via BioBERT) with drug databases (RxNorm/WHO-DD) to validate extracted terms.
- **Modular and Adaptable Pipeline:** Design an end-to-end architecture combining vision and language models tailored for semi-structured prescriptions.

Conclusion

Expected Advantages:

- Enhanced recognition capability for complex medical terminology [1]
- Improved performance in practical mobile implementations [5]
- Efficient processing compared to whole-document methods [4]

Future Work:

- Expanding support for additional regional languages [9]
- Integration with hospital electronic health record systems
- Edge-based processing for privacy and reduced latency
- Few-shot learning approaches to reduce annotation requirements

Practical Impact: Potential reduction in prescription interpretation errors, addressing a critical healthcare challenge that contributes to thousands of preventable adverse drug events annually [1]

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